NeuroMEP – Questions for the Engineering Team

Prepared 2 November 2026 for the MD and the Senior Electrical Engineer · WBS 1.5.5 engineering handover (due 4 November 2026)

NeuroMEP sizes every breaker and cable automatically. The software is built and tested, but some engineering values must come from you. Until they do, every result is marked **Provisional** and cannot be issued as Final.

We have studied the files you shared (the component spec, the CableSizer v1.1 workbook, the DB schedules, the manual and both presentations). Some questions are now answered; the ones below are still open. For each question we show **what we have now**, **why it matters** and **what we need from you**. A short answer per question is enough.

Part A – Needed before the calculation engine is built (5–6 Nov)

Q1. Which cable current-rating tables should we use?

What we have: the Ducab tables inside CableSizer v1.1 (based on BS 7671:2001).

Why it matters: these tables decide the size of every cable. They are over 20 years old and have never been checked against current Ducab catalogues or BS 7671 tables. A wrong value gives an undersized cable (overheating, fire risk) or an oversized one (wasted cost).

What we need: the current Ducab catalogue (or the BS 7671 tables you want used) for each cable type we offer: PVC / XLPE / fire-rated, Cu / Al, armoured / unarmoured, 1–4 core. Please also tell us which table each dropdown choice should use.

Q2. What demand factors should be the defaults?

What we have: example values from the Dubai Promenade project. They change with the board level:

Board	Lighting	Sockets	HVAC	Water heater	Laundry	Kitchen	Spare
DB	1.0	0.8	1.0	0.8	1.0	0.7	1.0
SMDB	–	0.6	1.0	0.4	0.4	0.6	0.8
MDB	–	–	–	–	–	–	0.8

Why it matters: demand factors decide the maximum demand, which decides every breaker and cable size upstream and the transformer size. Today NeuroMEP uses 1.0 (no reduction), which is safe but oversizes everything.

What we need: (a) May we use the table above as company defaults? (b) Do factors depend on the building type (residential, office, retail, landlord), and how should they combine when one panel serves several types? (c) Should each board apply its factor to the connected load (as CableSizer does)?

Q3. Which breaker catalogue should we use?

What we have: standard ratings, fault ratings per type (MCB 6–25 kA, MCCB 25–65 kA, ACB 50–85 kA) and a Merlin Gerin magnetic-trip / disconnection-time table.

Why it matters: without the manufacturer's data (Icu, Ics, Icm, let-through energy I^2t, tripping curves, derating at 50 °C) we cannot prove that a breaker clears a fault in time or that the cable survives the fault. Those checks show "Device data required" today.

What we need: the breaker catalogue you use (brand and range), and the limit up to which MCBs are used (we assume 63 A for now).

Q4. Which site conditions apply in Dubai?

What we have: CableSizer uses air 50 °C and ground 35 °C. The DEWA 2017 regulations state air 48 °C and ground 40 °C, soil 2.0 K·m/W.

Why it matters: a hotter ground means buried cables carry less current. Using 35 °C instead of 40 °C can give buried cables that are too small.

What we need: confirm the values to use: 50 / 35 °C (CableSizer) or 48 / 40 °C (DEWA 2017).

Q5. Which editions of the standards apply?

What we have: CableSizer follows BS 7671:2001 (16th edition), IEC 781 and IEC 364, which are all superseded.

Why it matters: formulas, limits and tables changed between editions. Authorities and clients check designs against the current editions.

What we need: the editions to apply (for example BS 7671:2018+A2, IEC 60909, DEWA regulations year).

Q6. Do you approve these corrections to the CableSizer method?

What we have: an independent check of CableSizer found some simplifications. NeuroMEP corrects them:

Item	CableSizer	NeuroMEP	Why
Fault voltage factor	c = 1.0	1.10 for maximum, 0.95 for minimum fault	c = 1.0 underestimates the maximum fault current
Voltage drop	fixed mV/A/m	resistance and reactance with the load power factor	more accurate for large cables
Earth-fault loop reactance	average of two sizes	phase plus protective conductor reactance	the average is not physically correct
Breakers inside panels	not derated	derated at 50 °C (manufacturer data)	panels are hotter than the room
Motor current	simple formula	includes motor efficiency	motors draw more than output kW suggests
Generator size	MD / 0.8	MD / 0.8 plus largest motor start, confirmed by manufacturer	motor starting can stall a generator
Values between table steps	nearest value	straight-line in between; blocked outside the table	avoids jumps and guessing

What we need: "approved", or the items you want changed.

Q7. When does an ELCB / RCD replace the earth-fault loop check?

What we have: the ELCB choices (30 mA / 100 mA ELCB, 0–300 / 0–500 mA ELS) and the rule "shown in the remarks column".

Why it matters: with an ELCB upstream, a circuit may be safe even when the loop impedance is high. Without the rule we either fail safe circuits or pass unsafe ones.

What we need: when the earth-fault loop check may be skipped because of an ELCB, and which ELCB ratings apply.

Part B – Needed for scenarios and reports (6–9 Nov)

Q8. Which loads go on the generator, and what is the Generator Load Analysis?

What we have: load categories (normal, essential, critical, fire & life safety). Nothing on what the Generator Load Analysis report should show.

Why it matters: this decides the generator size and the power-failure and fire-mode checks. It is also the one report you asked to be editable.

What we need: which loads are supplied in power failure and in fire mode, and the layout of the Generator Load Analysis (columns, totals, what can be edited).

Q9. Transformer, generator and busbar data

What we have: IEC typical values (for example transformer %Z 6 %, generator X″d 14 %).

Why it matters: fault levels, breaker ratings and busbar checks depend on the actual equipment. Busbars show "Datasheet required" today.

What we need: standard transformer and generator sizes with %Z / X″d, and a busbar datasheet (ratings,

resistance, reactance, short-time withstand Icw / Ipk, aluminium and sizes below 800 A).

Q10. Fault levels and clearing times

What we have: upstream fault level 350 MVA (CableSizer example) and the Merlin Gerin disconnection-time table.

Why it matters: the upstream fault level sets every fault current; the minimum fault level and the MV protection time decide whether faults are cleared in time.

What we need: the DEWA upstream fault level (maximum and minimum), the MV protection clearing time for the transformer-to-MDB cable, and whether the Merlin Gerin table may be used for all breakers.

Q11. Interim values – please confirm or correct

Value	We use	Why it matters
Transformer loading limit	80 %	when a second transformer is needed
Cable length allowance	10 % (plus 1 m per cable)	voltage drop and cable cost
Maximum cables in parallel	4 runs	when to move to a busbar
Load power factor by type	lighting 0.85, sockets / HVAC 0.8, heater 1.0 (CableSizer)	current and voltage drop
Target power factor (PFC)	0.92 (CableSizer)	capacitor bank size
Tray spare allowance	50 % (CableSizer)	tray width
Earthing system (TN-S / TN-C-S?)	not set yet – please tell us	earth-fault loop path
Motor starting current	6.5 × full load (CableSizer)	motor fault contribution and generator sizing

Q12. Fire-rated cables

What we have: fire & life-safety loads must have fire-rated cable on the generator side.

Why it matters: fire-rated cable is expensive; requiring it on the transformer side too changes cost and design.

What we need: must the transformer side of fire & life-safety loads also be fire-rated?

Part C – From the Technical Lead / MD

Q13. A reference project to test against

What we have: the Dubai Promenade workbook, but it was saved with broken links to an external file, so many results show #N/A.

Why it matters: acceptance testing (WBS 1.7.5 and 1.8) compares NeuroMEP with a design your team has already checked and issued. Without one we cannot sign the prototype off.

What we need: one hand-checked project with its inputs and issued schedules (a clean copy of Dubai Promenade with its linked DB file would work).

Q14. Output templates and symbols

What we need: the Excel schedule template with the company title block; symbols for equipment missing from the legend (transformer, generator, ATS, MDB, busbar, UPS, PFC…) and approval of our draft symbols; whether DXF drawings are needed.

Q15. Other confirmations

#	Question	What we assumed

a	Riser / transformer cost in AED for the Building Summary	shown as "–"
b	Does a riser feed boards directly, with the tap-off box implied?	yes
c	How are the ATS and two sources (transformer + generator) drawn?	by cables, normal and standby inputs
d	Area loads (m² × W/m²) from Excel: spread over 3 phases?	yes, equally
e	Phase balancing automatic with manual override?	yes
f	Is real-time collaboration (several users editing at once) needed for the prototype?	not planned

Summary – most urgent

Q1 cable tables, **Q2** demand factors, **Q3** breaker catalogue, **Q4** site temperatures and **Q13** reference project. Without them NeuroMEP still works, but no result can be issued as Final.

Reply against each question number. Short answers or attached files are fine.